

CANDIDATE DESIGN CHALLENGE

Frontend + UX Design Challenge

Industrial Cybersecurity Platform — Dashboard & Attack Path Experience

Purpose of this challenge

We want to understand how you think when designing a complex, data-heavy security product. This is not a branding exercise and not a request to reproduce an existing product. We are evaluating product thinking, information architecture, interaction design, visual quality, accessibility, and the ability to turn technical complexity into clear workflows.

Primary palette direction: ORANGE / BLACK / WHITE

Supporting highlight colours may be selected by the candidate where they improve clarity and accessibility.

Confidentiality note

This brief intentionally describes the product at a functional level only. Internal architecture, proprietary methods, detection logic, and implementation details are out of scope.

1. Product Context

You are designing two high-value surfaces for an industrial / operational-technology cybersecurity platform. The platform helps security and operations teams understand what exists in a networked environment, what is risky, what changed, and how risk may propagate through connected assets.

The experience must work for users with different goals:

- Security leadership / CISO — needs posture, priorities, trend and business impact.
- SOC / security analyst — needs triage, evidence, context and investigation speed.
- OT security analyst — needs asset, protocol, zone and operational context.
- Site / plant operator — needs clear impact, affected systems and operational relevance.

Design challenge

Create a product experience that is visually distinctive, but never sacrifices readability, trust or analyst speed. The interface should feel appropriate for an enterprise security product used in real operational environments.

2. Scope of the Challenge

Design the following two experiences:

1. A. Main Dashboard — the primary operational landing page.
2. B. Attack Path Map — an investigation workspace showing possible paths from an initial exposure or source to important target assets.

You may introduce supporting drawers, modals, side panels, filters and detail states where required to explain the experience. Do not design the full product.

3. Dashboard — Product Goal

Within approximately 5–10 seconds, the dashboard should help a user answer:

- Is the environment healthy and being monitored correctly?
- Is anything urgent happening now?
- What changed recently?
- Which assets, sites, zones or risks deserve attention first?
- Are critical systems exposed?
- Is there a meaningful attack path or movement opportunity that should be investigated?

3.1 Required Dashboard Information Areas

Area	What the user should understand
Executive / posture summary	High-level security posture, total assets, active/critical assets, high-priority findings, exposed critical assets, attack-path count or equivalent exposure summary.
Risk & findings	Severity distribution, major unresolved risks, top affected areas, risk trend, recently elevated issues.
Asset visibility	Asset inventory summary, new assets, unidentified or incomplete assets, inactive/offline assets, device-type/vendor/protocol distribution where useful.
Attack-path preview	High-risk path summary, important targets reachable, cross-zone exposure, top path to investigate, link to full attack-path view.
Network / topology insight	Zones/segments, communication hotspots, unusual concentration, critical relationships, high-level connectivity insight.
Recent change timeline	New assets, asset state change, new finding, risk change, communication change, major platform warning, operator action.
Platform / sensor health	Data recency, collection status, sensor health, visibility gaps, degraded state, dropped/limited data indicators where applicable.

3.2 Dashboard Interaction Requirements

- Global site / plant / area / zone filtering.
- Time range selection.
- Severity and criticality filtering.
- Search entry point for assets or findings.
- Drill-down from every important KPI or chart.
- Hover / focus detail where appropriate.
- Clear empty, loading, error and degraded-data states.
- Layout that remains useful for a small site as well as a large multi-site deployment.
- Visible distinction between “no issue exists” and “we do not currently have enough data.”

3.3 Dashboard Design Question

Candidate decision

You decide what belongs above the fold, what should be secondary, and which information should be visualised versus listed. We are specifically evaluating prioritisation—not how many widgets can fit

on one screen.

4. Attack Path Map — Product Goal

The Attack Path Map should help an analyst understand how an attacker or risky condition could traverse the environment from one asset or exposure point to a high-value target. The map should function as an investigation workspace, not just a decorative graph.

4.1 Core Path Elements

- Source / entry asset or starting condition.
- Intermediate pivot assets or relationships.
- Target / crown-jewel asset.
- Directed relationships between nodes.
- Protocol, service, trust or dependency context.
- Path risk / severity / confidence.
- Reason why each node or edge contributes to the path.
- Evidence supporting the path.

4.2 Node Design Requirements

A node should make the most important context visible without forcing the analyst to open a detail panel. Consider how to communicate:

- asset name / identity
- asset type or role
- criticality
- risk or exposure
- zone / location
- online or relevant operational state
- source / pivot / target role in the selected path
- uncertainty or confidence where relevant

4.3 Edge Design Requirements

- direction
- relationship type
- protocol / service
- risk relevance
- normal versus suspicious / risky relationship
- reason the edge can contribute to movement
- confidence / supporting evidence if useful

4.4 Investigation Panel

Selecting a node, edge or path should open context without losing the analyst's position in the graph. The detail experience should account for:

- summary
- why it matters
- supporting evidence
- relevant vulnerabilities / weaknesses
- related findings
- protocol / service context
- zone / asset context
- related paths
- recommended investigation / mitigation steps
- links to deeper asset or finding views

4.5 Attack Path Controls

- zoom, pan and fit-to-view
- search within graph
- filter by severity, confidence, site, zone, source or target
- single-path versus multi-path view
- collapse / expand groups or clusters
- highlight critical / shortest / selected path
- show reachable assets / blast radius
- compare paths where useful
- legend explaining node and edge semantics
- mini-map or navigation aid if the graph becomes large

Key UX problem

A graph can look impressive and still be unusable. Your solution should demonstrate how a user can understand a complex path quickly, isolate what matters, and avoid being overwhelmed by hundreds of nodes and edges.

5. Functional States to Consider

State	Expected UX consideration
Normal	Healthy data, no urgent issue, current information.
High-risk	Critical/high finding or attack path needs attention.

State	Expected UX consideration
Degraded data	Sensor, collection or visibility quality is impaired. Users must know conclusions may be incomplete.
Unknown / uncertain	Insufficient information. Do not visually imply certainty.
Empty	No findings, no path, no data, or no result after filtering.
Large data volume	Thousands of assets or relationships. Design must remain navigable.
Partial data	Some asset attributes, relationships or evidence are missing.
Action feedback	Acknowledged, assigned, reviewed, suppressed or other action states should have clear feedback.

6. Design Quality & International Standards

The design should be suitable for an international enterprise cybersecurity product. At minimum, demonstrate awareness of:

- WCAG 2.2 accessibility principles, including colour contrast and non-colour-only status communication.
- Keyboard and focus-state awareness for primary interactions.
- Readable typography and hierarchy at dense information levels.
- Consistent interaction patterns, spacing and component behaviour.
- Responsive thinking for common desktop resolutions; desktop-first is acceptable for the submitted challenge.
- Clear confirmation for destructive or high-impact actions.
- Tooltips / help only where they reduce ambiguity; avoid hiding essential meaning behind hover-only interactions.
- Status, severity and confidence should not be represented by colour alone.
- Charts and graph elements should have meaningful labels, legends or alternate cues.

7. Visual Direction

Required base palette:

ORANGE

BLACK

WHITE

You may introduce supporting colours for severity, success, information, selection and graph semantics. You decide the visual language: typography, cards, graph style, spacing, iconography, light/dark balance and component treatment.

We are not looking for

A generic SOC dashboard copied from common templates, excessive neon “hacker” visuals, gratuitous 3D effects, or a UI that depends on decoration rather than hierarchy and workflow.

8. Required Deliverables

3. Dashboard — high-fidelity desktop design.
4. Attack Path Map — high-fidelity desktop design.
5. At least one node/edge/path detail state or investigation side panel.
6. Key interaction states or a lightweight clickable prototype.
7. Brief design rationale (1–2 pages or equivalent frames).
8. A small component / token view showing how the design remains consistent.

Preferred submission: Figma link. A PDF may also be included for presentation.

8.1 Design Rationale Must Explain

- What information you prioritised and why.
- How the dashboard serves both leadership and analyst workflows.
- How you prevent attack-path visual overload.
- How the design handles uncertainty and incomplete data.
- How the design scales to large environments.
- Any assumptions made because information was intentionally not provided.

9. Timebox & Expectations

We care more about judgement than pixel volume. A focused solution with strong reasoning is preferable to a large number of loosely connected screens. The candidate may state the time spent and which areas would be developed further with more time.

10. Constraints

- Do not attempt to reconstruct or reverse-engineer proprietary product internals.
- Do not design the entire application.
- Do not assume perfect or complete data.
- Do not assume every user is a cybersecurity expert.
- Do not rely on colour alone for severity or state.
- Do not make every metric equally prominent.
- Do not make every graph element permanently visible if progressive disclosure would improve usability.

11. Evaluation Criteria

Area	Weight	What we look for
Product understanding	20%	Understands the problem, users and workflows; avoids superficial dashboard thinking.
Information architecture	20%	Prioritises information well and creates a clear hierarchy.
Interaction design	20%	Drill-downs, graph interactions, filtering, states and actions are coherent.
Visual design	15%	Distinctive, polished, consistent and appropriate for enterprise security.
Attack-path usability	15%	Graph remains understandable, investigable and scalable.
Accessibility & resilience	10%	Handles accessibility, uncertainty, partial/degraded data and edge states.

12. Optional Bonus

- Responsive adaptation concept.
- Dark/light theme logic.
- Keyboard navigation / accessibility notes.
- Motion or graph-transition rationale.
- A compact notification / triage pattern.
- A “degraded visibility” experience showing how the product communicates uncertain monitoring coverage.

13. Submission

Please submit the design together with your short rationale. We may use the follow-up discussion to ask why particular elements were prioritised, how your graph would behave at larger scale, and how you would revise the design after real user feedback.

Final note

There is no single “correct” visual answer. We are looking for strong product judgement, clear reasoning, deliberate trade-offs, and a design that could realistically evolve into a high-quality

enterprise product.